## Activity

Here is an interesting number game to thrill your friends. Follow the steps.

1. Ask your friend to think a number in mind. Let the number be in between 1 to 10. (for example, let the number be 5).
2. Double the number and add to its square. (So, 10 + 25 = 35).
3. Add one to the result. (So, 35 + 1 = 36).
4. Ask the final resultant number.
5. The number will be a perfect square. Consider its base in the exponent form and reduce 1 from it. The result is the number in your friend's mind. ( $36 = 6^2 \Rightarrow 6$ is the base; so, $6 - 1 = 5$ ).

   Remember the perfect square numbers and their corresponding base numbers $1 \rightarrow 1$, $4 \rightarrow 2^2$, $9 \rightarrow 3^2$, $16 \rightarrow 4^2$, $25 \rightarrow 5^2$, $36 \rightarrow 6^2$, $49 \rightarrow 7^2$, $64 \rightarrow 8^2$, $81 \rightarrow 9^2$, $100 \rightarrow 10^2$

## Exercise 3.1

1. Fill in the blanks.

   (i) $(p-q)^2$ = ----------.

   (ii) The product of $(x + 5)$ and $(x - 5)$ is ----------.

   (iii) The factors of $x^2 - 4x + 4$ are ----------.

   (iv) Express $24ab^2c^2$ as product of its factors is ---------.

2. Say whether the following statements are True or False.

   (i) $(7x + 3)(7x - 4) = 49x^2 - 7x - 12$ .

   (ii) $(a-1)^2 = a^2 - 1$.

   (iii) $(x^2 + y^2)(y^2 + x^2) = (x^2 + y^2)^2$ .

   (iv) $2p$ is the factor of $8pq$.

3. Express the following as the product of its factors.

   (i) $24\ ab^2c^2$.

   (ii) $36\ x^3y^2z$.

   (iii) $56\ mn^2p^2$.

4. Using the identity $(x + a)(x + b) = x^2 + x(a + b) + ab$, find the following product.

   (i) $(x + 3)(x + 7)$ (ii) $(6a + 9)(6a - 5)$

   (iii) $(4x + 3y)(4x + 5y)$ (iv) $(8 + pq)(pq + 7)$



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